Exploiting conceptual dictionary for semantic textual similarity

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***Abstract*—Semantic textual similarity (STS) aims to measure the degree of semantic equivalence between two text units. It plays an important role in many natural language processing tasks, including information retrieval, question answering, and text summarization. In this paper, we propose a knowledge-based approach for computing semantic similarity between sentences using a conceptual dictionary.**

The proposed method extracts lemmas from input sentences and retrieves their associated concepts from the Slovak Conceptual Dictionary developed by the Kempelen Institute of Intelligent Technologies. Based on the extracted concepts, a semantic tree structure is dynamically constructed for each pair of sentences. Semantic similarity between concepts is computed using the Wu–Palmer similarity measure, and the final sentence similarity score is obtained using a best-match aggregation strategy.

The proposed approach was evaluated on a subset of the Slovak SICK dataset (SICK-SK) containing 7569 sentence pairs. The quality of the computed similarity scores was assessed using the Pearson correlation coefficient between model predictions and human similarity annotations.

The experimental results achieved a Pearson correlation of 0.3094, indicating a moderate positive correlation between the proposed approach and human judgments. This shows improvement compared to existing knowledge-based algorithms applied to Slovak texts. The results demonstrate that knowledge-based approaches relying on structured lexical resources can provide interpretable and lightweight solutions for semantic similarity computation.

***Index Terms*—semantic textual similarity, knowledge-based methods, conceptual dictionary, Wu-Palmer similarity, natural language processing**

I. Introduction

Semantic similarity expresses the degree of semantic closeness between two text units, such as words, phrases, sentences, paragraphs, or entire documents. It can be applied to texts of different lengths, meaning that the compared texts may contain different numbers of words or phrases. Its purpose is to determine the extent to which two pieces of text convey the same or a similar meaning. Unlike simple lexical comparison, semantic similarity focuses on understanding the meaning behind words rather than only comparing their surface forms.

When evaluating semantic similarity, the concepts represented by words are compared. For example, the terms “dog” and “cat” are closer in meaning than “dog” and “car”, because both belong to the category of animals. The result of a semantic similarity computation is usually a numerical value expressing the degree of similarity between the compared texts. This value often ranges from 0 to 1, where 0 represents complete semantic dissimilarity and 1 represents complete semantic similarity. In practice, however, other ranges or categorizations can also be used, dividing the level of similarity into several degrees, from no similarity to complete semantic equivalence.

The importance of semantic textual similarity (STS) lies in its wide range of applications in natural language processing. It plays an important role in systems such as automatic text summarization, machine translation, plagiarism detection, and intelligent chatbots that need to understand different formulations of the same question.

Semantic similarity can be determined in various ways, ranging from classical lexical approaches to modern statistical methods and deep learning models. Another important group of approaches are methods based on external knowledge sources, which exploit structured semantic relations between concepts.

In this work, we propose a knowledge-based approach for measuring semantic similarity between sentences using a conceptual dictionary. The proposed method extracts lemmas and their associated concepts from the compared sentences and uses this information to construct a semantic representation of the input texts. Instead of relying on a predefined global taxonomy, the model dynamically constructs a semantic tree for each pair of sentences based on the concepts identified in the input.

The similarity between individual concepts is computed using the Wu–Palmer similarity measure, which evaluates

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semantic closeness according to the hierarchical relations between concepts. The final similarity score between sentences is obtained using a best-match strategy that aggregates the similarities of individual concepts from both sentences.

The proposed approach aims to provide an interpretable and lightweight method for sentence similarity computation that relies on explicit semantic relations rather than large amounts of training data. By exploiting structured knowledge from a conceptual dictionary and dynamically constructing semantic structures for each sentence pair, the model focuses on the most relevant conceptual relations when evaluating semantic similarity. The main contribution of this work is a knowledge-based sentence similarity model that dynamically constructs semantic trees from a conceptual dictionary and evaluates similarity using the Wu–Palmer measure.

II. Background

Knowledge-based methods determine semantic similarity based on relationships between concepts in a semantic network. Semantic networks are structures consisting of nodes representing objects, concepts, or events, and edges expressing relationships between them. An important feature of semantic networks is that important information about a particular concept can be derived from the nodes that are directly connected to it, without the need to search the entire system.

These methods use existing linguistic knowledge, such as hierarchical superordinate and subordinate relationships (is-a), synonymy, and polysemy. The similarity of two concepts is then determined by their location in the semantic network structure, for example, by the length of the path connecting them or by the depth of their closest common superordinate concept.

Based on the method of calculation, these methods are divided into node-based, which use the information content of concepts, edge-based, which measure the distance between nodes, and hybrid, which combine both approaches. Among the best-known knowledge-based methods are the Wu-Palmer, Shortest Path, and Leacock Chodorow algorithms.

A. Wu–Palmer Similarity

The Wu–Palmer [1] algorithm determines semantic similarity between two concepts based on their depth in a semantic hierarchy and the depth of their closest shared ancestor, referred to as the Least Common Subsumer (LCS). The deeper this common ancestor is in the hierarchy, the more semantically similar the two concepts are considered to be.

The similarity is computed using the following formula:

$$sim(s_1, s_2) = \frac{2 \cdot depth(LCS)}{depth(s_1) + depth(s_2)} \tag{1}$$

where $depth(x)$ represents the depth of concept x in the hierarchy. The resulting score takes values in the range $0 < sim \leq 1$. The value cannot be zero because the depth of the LCS is never zero, as the root of the hierarchy always has depth equal to one.

B. Shortest Path

The Shortest Path [2] algorithm determines semantic similarity between two concepts based on the length of the shortest path that connects them in a semantic network. In such a network, each concept is represented as a node and the relations between concepts are represented as edges. The shorter the path between two nodes, the closer the concepts are in meaning.

The semantic distance between two concepts is defined as:

$$dist(S, T) = \min_{s \in S,\, t \in T} path_length(s, t) \tag{2}$$

where S and T represent sets of meanings (synsets) of the two concepts. Based on this distance, semantic similarity can be expressed as:

$$sim_{SP}(c_1, c_2) = \frac{1}{1 + dist(c_1, c_2)} \tag{3}$$

The resulting score takes values in the range $0 < sim \leq 1$, where higher values indicate greater semantic similarity.

C. Leacock–Chodorow Similarity

The Leacock–Chodorow [3] algorithm is an extension of the Shortest Path method, as it considers not only the length of the shortest path between two concepts but also the overall depth of the taxonomy. It therefore represents a normalized measure of similarity that takes into account the position of the compared concepts within the hierarchy.

The similarity is computed as the negative logarithm of the ratio between the shortest path between two concepts and twice the maximum depth of the taxonomy:

$$sim_{LCH}(c_1, c_2) = -\log\left(\frac{spath(c_1, c_2)}{2D}\right) \tag{4}$$

where $spath(c_1, c_2)$ represents the shortest path between concepts c_1 and c_2, and D denotes the maximum depth of the taxonomy. The shorter the path between the concepts and the deeper their common ancestor in the hierarchy, the higher the resulting similarity score.

III. Conceptual Dictionary

The proposed approach relies on the Slovak Conceptual Dictionary developed by the Kempelen Institute of Intelligent Technologies (KInIT) [4]. This resource provides structured semantic information about words and their meanings and organizes lexical knowledge in the form of interconnected concepts and semantic relations between them.

In a conceptual dictionary, each word is associated with a concept that represents its meaning. Concepts are organized within a hierarchical structure that captures semantic relations between them. This hierarchy enables the identification of shared ancestors between concepts and allows the computation of semantic similarity based on their relative positions in the structure.

The dictionary also contains relations between concepts, such as synonym relations and hierarchical relations. These

relations allow the creation of semantic networks that represent connections between concepts and support semantic analysis of text.

The conceptual dictionary used in this work is accessible through a publicly available API[1]. Through this interface it is possible to retrieve structured information about words occurring in input sentences, including their lemmas, associated concepts, and semantic relations between concepts. These data are used in the proposed approach to construct semantic structures that serve as the basis for sentence similarity computation.

The Slovak Conceptual Dictionary represents one of the largest structured lexical resources available for the Slovak language and is designed to support semantic processing tasks in natural language processing applications.

IV. Our Approach

The proposed approach computes semantic similarity between sentences using structured knowledge obtained from a conceptual dictionary. The approach transforms the input sentences into a semantic representation and subsequently evaluates their similarity based on relations between the extracted concepts. The overall process consists of several steps, including lemma extraction, concept retrieval from the conceptual dictionary, construction of a semantic tree, computation of concept similarity, and aggregation of these similarities into a final sentence similarity score. The overall workflow of the proposed approach is illustrated in Fig. 3. The model processes input sentences through several stages including lemma extraction, concept retrieval, semantic tree construction, and sentence similarity computation.

A. Lemma Extraction

The first step of the proposed method is the extraction of lemmas from the input sentences. Lemmatization reduces words to their base forms, which represent the canonical form of a word. This step is important because different inflected forms of a word may correspond to the same underlying concept. For each sentence, the words are processed and transformed into their corresponding lemmas. Only lemmas that can be associated with concepts in the conceptual dictionary are considered in further processing. Words that do not have a corresponding concept representation are excluded from the model, since they cannot be incorporated into the semantic structure used for similarity computation.

B. Concept Retrieval

After lemma extraction, the associated concepts are retrieved from the conceptual dictionary through the available API. Each lemma is mapped to one or more concepts representing its meaning within the lexical resource. The retrieved information includes the concept identifier and the semantic relations connected to the concept. These concepts form the basic elements used to construct the semantic representation of the compared sentences.

[1]https://pojmy.kinit.sk/concepts

C. Semantic Tree Construction

The semantic structure used in the proposed approach is constructed dynamically for each pair of sentences. Instead of using the entire taxonomy of the conceptual dictionary, the model creates a local semantic tree that contains only the concepts relevant to the compared sentences. The tree structure begins with a root node, under which intermediate layers representing concept namespaces are created. Individual concepts extracted from the sentences are then inserted as nodes in the structure. Synonym relations between concepts are used to extend the structure and connect semantically related concepts. By constructing the semantic tree dynamically, the model focuses only on the concepts that appear in the processed sentences, which reduces the size of the structure and improves computational efficiency.

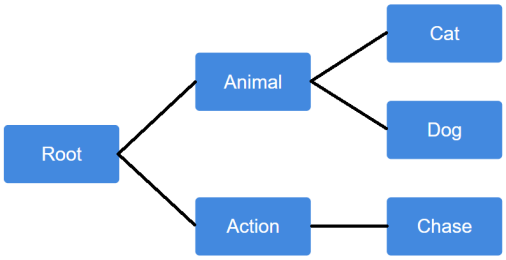

Fig. 1. Example of a dynamically constructed semantic tree created from concepts extracted from input sentences.

D. Concept Similarity Computation

Once the semantic tree is constructed, similarity between individual concepts is computed using the Wu–Palmer similarity measure. This measure evaluates the semantic closeness of two concepts according to their position within the hierarchical structure. The similarity value reflects how closely two concepts are related based on the depth of the concepts and the depth of their lowest common subsumer within the hierarchy.

E. Sentence Similarity Computation

The final similarity between sentences is derived from the similarities between individual concepts. The model applies a best-match strategy that allows comparison of sentences with different lengths. For each concept extracted from the first sentence, the similarity is computed with all concepts from the second sentence. The highest similarity value is selected as the best match. This process is repeated for all concepts in the first sentence. The overall sentence similarity score is then computed as the arithmetic mean of the best similarity values obtained for each concept. If no similarity can be computed between the concepts of the compared sentences, the resulting similarity score is set to zero.

V. Experiments and Evaluation

To evaluate the proposed approach, experiments were conducted using a Slovak version of the SICK [5] dataset, referred

Algorithm 1: Sentence Similarity using Best-Match Strategy
Input: lemma sets L_1, L_2, semantic tree T
Output: similarity score $sim(S_1, S_2)$

```
scores = [ ]
for each lemma w1 in L1 do
    best = 0
    for each lemma w2 in L2 do
        score = WuPalmer(w1, w2, T)
        if score > best then
            best = score
    append best to scores
if scores is empty then return 0
return
```

$$sim(S_1, S_2) = \frac{1}{|scores|} \sum scores$$

Fig. 2. Best-match computation of sentence similarity

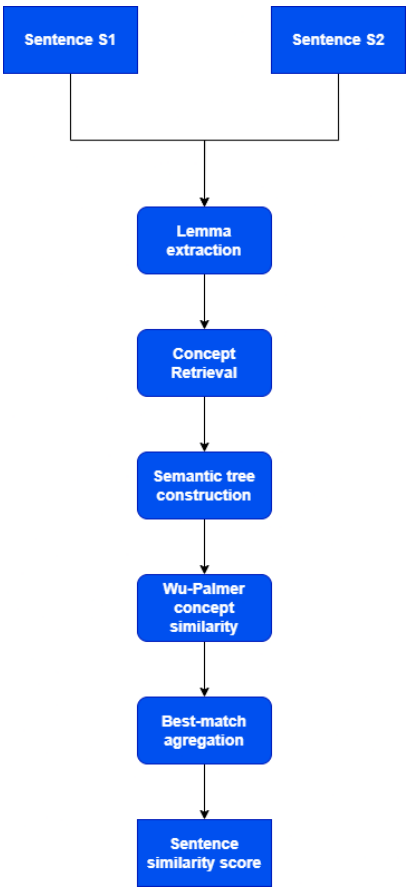

Fig. 3. Overview of the proposed sentence similarity computation process.

to as SICK-SK. The SICK (Sentences Involving Compositional Knowledge) dataset is a widely used benchmark for evaluating semantic textual similarity systems. It contains pairs of sentences annotated with similarity scores provided by human annotators.

Each record in the dataset consists of two sentences and a numerical similarity score representing the degree of semantic equivalence between them. The scores typically range from 1 to 5, where higher values indicate greater semantic similarity.

For the purposes of this experiment, a subset of the SICK-SK dataset was used, containing 7569 sentence pairs. Each pair consists of two Slovak sentences together with a human similarity score.

During the experiment, each sentence pair was processed using the proposed similarity model. First, lemmas were extracted from the sentences using the conceptual dictionary API. Only lemmas associated with concepts in the conceptual dictionary were considered in further processing. For each pair of sentences, the retrieved concepts were used to dynamically construct a semantic tree structure that represents semantic relations between the concepts.

After constructing the semantic tree, semantic similarity between concepts from the two sentences was computed using the Wu–Palmer similarity measure. For each concept from the first sentence, the model identified the most similar concept from the second sentence using a best-match strategy. The final similarity score between the two sentences was computed as the arithmetic mean of these best similarity values.

The computed similarity scores produced by the model were then compared with the human similarity annotations provided in the dataset. To measure the agreement between the two sets of scores, the Pearson correlation coefficient was used.

The Pearson correlation coefficient is defined as

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{5}$$

where x_i represents the similarity score produced by the proposed approach and y_i represents the similarity score assigned by human annotators. The values $\bar{x}$ and $\bar{y}$ denote the mean values of the respective variables.

The Pearson correlation coefficient takes values in the interval $[-1, 1]$, where values closer to 1 indicate a strong positive correlation between the compared variables.

The evaluation of the proposed approach on the SICK-SK dataset resulted in a Pearson correlation coefficient of

$$r = 0.3094 \tag{6}$$

which indicates a moderate positive correlation between the similarity scores produced by the proposed approach and the similarity judgments provided by human annotators. In comparison, traditional knowledge-based methods such as the Wu-Palmer similarity applied to the Slovak WordNet achieve a Pearson correlation of r = 0.237 on the same dataset [6]. This represents improvement in capturing semantic nuances in the Slovak language with knowledge-based methods.

VI. Conclusion and Future Work

This paper presented a knowledge-based approach for computing semantic similarity between sentences using the Slovak Conceptual Dictionary. The proposed method transforms input sentences into a semantic representation by extracting lemmas and retrieving their associated concepts from the conceptual

dictionary. Based on these concepts, a semantic tree is dynamically constructed and concept similarity is computed using the Wu–Palmer measure. The final sentence similarity score is obtained using a best-match aggregation strategy.

The proposed approach was evaluated on a subset of the Slovak SICK dataset containing 7569 sentence pairs. The evaluation using the Pearson correlation coefficient resulted in a value of 0.3094, indicating a moderate positive correlation between the similarity scores produced by the model and human annotations. These results demonstrate that knowledge-based approaches based on structured lexical resources can provide interpretable solutions for semantic textual similarity tasks in Slovak.

Future work will focus on extending the proposed model by incorporating additional semantic relations from the conceptual dictionary and improving the construction of the semantic tree. Another direction of future research is the improvement of the best-match aggregation strategy used for computing sentence similarity. Alternative matching or aggregation methods may be explored to better capture semantic relations between concepts. Moreover, we intend to explore incorporating large language models for tree construction enhancement.

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